AYOMIDE ABILEWA

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Research Profile

Most of the work so far has gone into measurement and detection under non-ideal conditions. Recent projects: a two-stage cascaded detector that fuses RGB and edge-enhanced models for pipe anomaly identification, a multi-sensor multispectral acquisition node, and control-systems practice on Quanser robotics hardware using MATLAB and Simulink.

Education

B.Sc. Electronic and Electrical Engineering 2021–2027 (expected)

Obafemi Awolowo University, Ile-Ife, Osun State, Nigeria

Relevant coursework: Control Systems, Measurement and Instrumentation, Digital Electronics, Signal Processing, Power Systems, Microprocessors and Embedded Systems, Electronic Circuit Design.

Research Interests

- Detection under degraded conditions: How to keep detection precision steady when lighting, surface condition and viewing angle vary within a single run, instead of tuning for one condition and losing the others.
- Measurement and calibration: Turning raw sensor output into a measurement that means something: calibration, separating quantities that are easy to conflate, and the signal standards industry already relies on.
- Feedback and control: Closed-loop behaviour on real hardware, from a thermistor-driven PWM loop to PID control on Quanser platforms, where the actuator changes the quantity being measured.
- Embedded and real-time vision: Running vision pipelines within real constraints: modest hardware, live video, and the requirement that the system respond now rather than after a batch job.
- Systems that degrade rather than fail: Designing explicit fallback behaviour so losing a network, a model or a sensor costs quality instead of function.

Research and Technical Projects

Live Video Pipe Anomaly Detection 2026–Present

Python, YOLOv8, YOLOv11, Weighted Boxes Fusion

- Building a two-stage cascaded detector for real-time pipe anomaly identification in industrial inspection.
- Two YOLO models run in parallel, one on RGB and one on edge-enhanced input, merged with Weighted Boxes Fusion to hold precision across lighting and surface conditions.

Multispectral Acquisition Device for Hydroponics 2025

ESP32, GY-2561, GY-31, I2C

- Built an ESP32 acquisition node measuring light intensity and colour spectrum with the GY-2561 and GY-31, streaming live to a Blynk dashboard.
- Two I2C sensors on one bus, with arbitration and calibration handled so the readings mean something in absolute terms.

AI-Powered Smart Attendance System Feb–Mar 2026

Python, OpenCV, YuNet, SFace

- Owned the vision pipeline and the Flask REST API: YuNet detection, SFace recognition, cosine-similarity matching.
- Tested against real lecture-hall lighting and off-axis faces, not curated frontal shots.
- Defended it at the final-year project review; invited afterwards to build a production version.

github.com/ayomide-abilewa/Smart-Attendance-System

Research and Engineering Experience

Research and Development Intern (SIWES) 2024

ACE Quanser Robotics Lab, Obafemi Awolowo University, Ile-Ife, Nigeria

- Operated three Quanser platforms in MATLAB and Simulink: the QDrone quadrotor, the QCar ground vehicle, the QArm 6-DOF manipulator.
- PID control and sensor feedback tuned on live hardware, not in simulation.
- Gave the lab orientation talk on the Quanser Aero 2: architecture, control principles, safety.

Facilities Engineering Management Intern (SIWES) Mar 2026–Present

Chevron Nigeria Limited, Lagos, Nigeria

- Rotating through the Electrical and Instrumentation (E&I) team on field instrumentation and electrical systems.
- Read P&IDs before fieldwork; traced control loop configurations, I/P converters and 3–15 PSI pneumatic signalling.

Embedded Systems and EEE Tutor 2023–Present

Astar Tutorials, Ile-Ife, Nigeria

Technical Skills

Programming: Python, C/C++, MATLAB, JavaScript, Verilog/HDL, Bash

Embedded and Hardware: Arduino, ESP32, STM32, Raspberry Pi, FPGA, I2C / SPI / UART, Sensor integration

AI and Machine Learning: YOLOv8/v11, OpenCV, TensorFlow (fundamentals), PyTorch (fundamentals), Weighted Boxes Fusion

Instrumentation and Control: P&ID interpretation, PID control loops, Pneumatic systems, I/P converters, 4–20 mA and 3–15 PSI standards

Tools: VS Code, Git/GitHub, MATLAB/Simulink, KiCad, Flask, React/TypeScript, PostgreSQL, Blynk IoT

Training and Certifications

FPGA Design and HDL/Verilog Workshop (SWEP), Faculty of Engineering, Obafemi Awolowo University 2024

Robotics Lab Orientation and Operations, ACE Quanser Lab, Obafemi Awolowo University 2024

Teaching and Academic Service

SPAW Project Lead and Syllabus Team Lead 2024–Present

IEEE OAU Student Branch

- Lead SPAW 3.0, a six-session embedded systems and entrepreneurship curriculum for secondary school students.